

Stagnone di Marsala digital twin: project resumption and status

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To: Prof. Giuseppe Ciraolo, Prof. Mauro De Marchis

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1. Summary

Work on the coupled three-dimensional model resumed on 5 August after a pause since mid-June, due to the rewriting of the manuscripts related to GEE. As we already had several runs done and the skeleton of the paper in place, the main outcome of the past three days was to compile everything and finalise the first draft. It is now complete in LaTeX, formatted for *Estuarine, Coastal and Shelf Science* and within that journal's limits, and provisionally titled **“Transport validation of a wave-coupled lagoon model: interacting effects of seagrass roughness and bed mobility”**, with the three of us as authors.

A numerical problem that had cost us two of the eight experiments was also traced and fixed today, and the corresponding re-runs are in progress (Section 3).

2. What the study asks and what it found

Coastal lagoon models are routinely validated against water level and then used to describe transport, so I set the paper to ask whether the first licenses the second. Six configurations were run over the same nine days of July 2025, covering the drifter deployments, from the same initial condition and forcing, differing only in wave coupling, bed mobility, and whether seagrass resistance is uniform or distributed from a satellite classification. Each was scored against three tide gauges and against 35 GPS drifters over twelve deployments.

Water level does not separate the configurations. Anomaly RMSE differs between them by six millimetres or less at any station, within the measurement uncertainty, and no configuration is consistently best. Over the same simulations Liu–Weisberg skill spans 0.505 to 0.655 and endpoint separation 334 to 661 m. A calibration on water level alone would have had no basis on which to prefer any of them.

No treatment improves Lagrangian skill on its own, whether seagrass roughness, bed mobility or wave coupling. Applied together, distributed roughness and bed mobility raise skill by 0.13 to 0.15, improving 32 of the 35 drifters, and only that combination reproduces the observed transport rate.

Splitting the trajectory error into magnitude and direction shows what each treatment corrects. This is the most recent result and the one I consider the paper's contribution.

Configuration	Path length (sim/obs)	Net displacement (sim/obs)	Heading error (deg)
Four fixed-bed members	0.77–0.80	0.78–0.82	15.8–16.2
Mobile bed, uniform roughness	0.94	0.95	12.5
Mobile bed, distributed roughness	0.98	0.99	8.4

The four fixed-bed members coincide in all three measures. Bed mobility acts on magnitude, restoring the distance travelled. Distributed roughness acts on direction, and only once the bed can move: on a fixed bed it leaves the heading error unchanged. I could not find a simpler explanation, having tested and rejected three, and the paper reports it as a narrowing rather than a complete mechanism.

Two further results support the modelling choices. The canopy parametrisation attenuates near-bed wave orbital velocity in proportion to canopy density, and surface speed exceeds bed speed by a factor of



1.9 to 4.5 at sub-metre mean depth, consistent with De Marchis et al. (2012) here and the quantitative case for solving in three dimensions.

3. A problem traced and fixed today

Two of the eight cells of the factorial, those combining a mobile bed with no wave coupling, had been aborting on the model's velocity cap. The manuscript as it stands reports this as a property of the system. That reading was wrong, and I am glad the check was made before submission.

The cause is the coupling between suspended sediment and density, which is the only route by which sediment can move momentum in a deep offshore cell holding no erodible material. It also explains an earlier fix that had failed. With the coupling disabled the run passed the abort point without incident, at nearly double the time step, and three re-runs are under way so that all four mobile-bed members share the setting. They should finish overnight.

The design then closes at the full eight cells, and the effect of wave coupling becomes measurable on a mobile bed, which the manuscript currently states cannot be done. Sections 3.6, 4.3, 5 and 6 will be revised accordingly. I do not expect the results in Section 2 to move, since they rest on members that were never affected, but they will be recomputed and checked.

4. Manuscript status and one question for you

The draft is complete across all six sections, with eight figures and three tables, 6200 words against a limit of 8000, and 24 references. Highlights and a cover letter are drafted. Outstanding is a graphical abstract, which *ECSS* requires.

The one point on which I would value your judgement is strategic rather than editorial: **whether to submit before obtaining a new bathymetric survey**. The configuration that best reproduces observed transport is the one with a mobile bed, and its critical shear stress for erosion has not been constrained against measured bathymetric change. We are using the old bathymetry, so it is possible that the mobile bed helps because it moves the old morphology towards the present one, in which case the improvement is a compensation rather than a physical result. The manuscript states this limitation openly, and notes that the lagoon bed changes by up to 1.0 m over nine days in isolated cells, which is not a plausible rate. If there is a realistic chance of obtaining new bathymetry soon, I would rather wait and rerun the simulations with it before submitting than defend the present version in review.

On the wider plan: after the holidays, and before leaving for Malta, my intention is to close this paper and begin drafting the remaining themes we discussed. Those are the effect of the multi-decadal *Posidonia* decline on wave attenuation, near-bed exposure and residence time; the operational forecast chain with the WetWise portal, together with the Isola Grande channel reopening scenario; and wave-driven sediment mobilisation with the Trapani canal contribution to lagoon siltation.

The full draft, figures and analysis code are in the project repository, and I can send the compiled PDF on request.

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